

Effects of drinking supplementary water at school on cognitive performance in children.



We investigated the beneficial effects of drinking supplementary water during the school day on the cognitive performance and transitory subjective states, such as fatigue or vigor, in 168 children aged between 9 and 11 years who were living in a hot climate (South Italy, Sardinia). The classes were randomly divided into an intervention group, which received water supplementation, and a control group. Dehydration was determined by urine sampling and was defined as urine osmolality greater than 800 mOsm/kg H₂O (Katz, Massry, Agomn, & Toor, 1965). The change in the scores from the morning to the afternoon of hydration levels, cognitive performance and transitory subjective states were correlated. In line with a previous observational study that evaluated the hydration status of school children living in a country with a hot climate (Bar-David, Urkin, & Kozminsky, 2005), our results showed that a remarkable proportion of children were in a state of mild, voluntary dehydration at the beginning of the school day (84%). We found a significant negative correlation between dehydration and the auditory number span, which indicates a beneficial effect of drinking supplementary water at school on short-term memory. Moreover, there was a positive correlation between dehydration and performance in the verbal analogy task. The results are discussed in the light of the complexity of the neurobiological mechanisms involved in the relationship between hydration status and cognition. Copyright © 2012 Elsevier Ltd. All rights reserved.